

Complex ET Reconstruction

The physics target is the complex transverse field integral. Strong and stressed configurations can be compared directly against the full-grid reference in both the complex plane and along z .

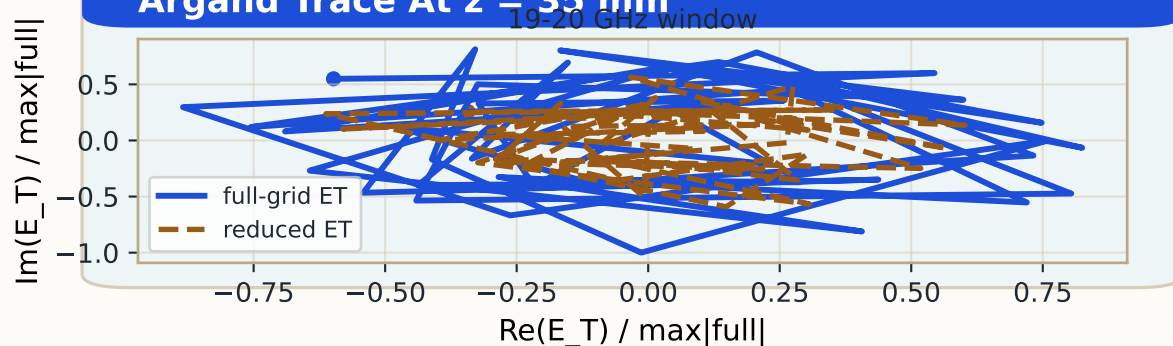
$$E_T(z, \nu) = \iint_A E(x, y, z, \nu) dx dy$$

The comparison below uses normalized ET so the phase-space shape remains visible even for the stressed configuration.

Representative Strong Multi-Gap Case

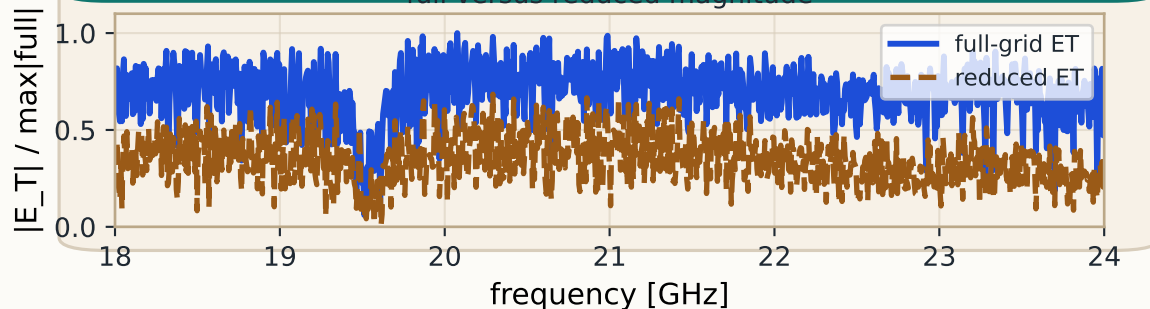
lines=1, z:slices=1, gaps=2; NRMSE = 37.33
retained z: 22, 35 mm; display z = 35 mm

Argand Trace At z = 35 mm



$|E_T|(\nu)$ At z = 35 mm

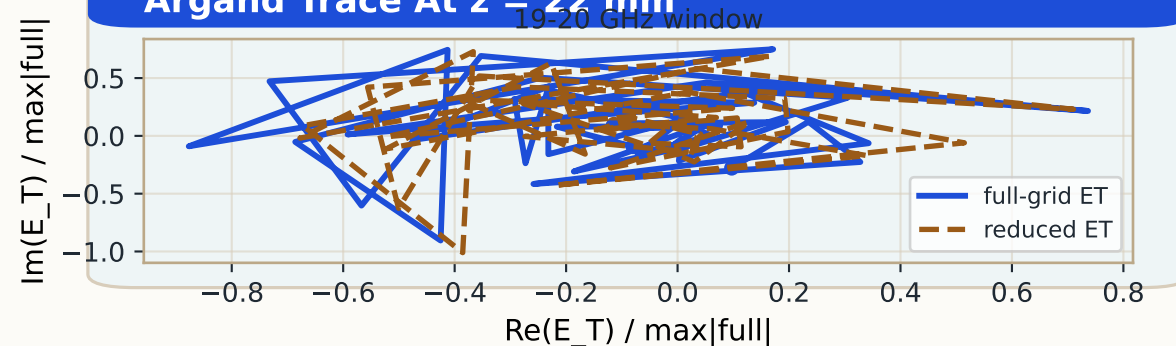
full versus reduced magnitude



Representative Stress test

lines=4, z:slices=3, gaps=4; NRMSE = 65.94
retained z: 2, 3, 4, 16, 17, ... mm;
display z = 22 mm

Argand Trace At z = 22 mm



$|E_T|(\nu)$ At z = 22 mm

full versus reduced magnitude

